

≡ Hide menu

Problem 1: Constructing Height Balanced Binary Search Tree

Problem 2: Two Sum Problem

Problem 3: Image Blurring Lab and Assignment

Lab Practice Lab: Image Edge Detection 1h

Graded Assignment: Graded Quiz: Solving Miscellaneous Problems Submitted

Wrap Up

Your grade: 100%

Next item →

Your latest: 100% • Your highest: 100% • 24 hours until you can retake this quiz

Graded Quiz: Solving Miscellaneous Problems

1. You are using GitHub Copilot for image blurring. Which of the following techniques would you NOT be able to implement using this Copilot?

5 / 5 points

Review Learning Objectives

Median filter blurring

☐

Gaussian blurring

☐

Box blurring

☐

None of the other options

☒

Assignment details

Due Jul 14, 11:59 PM IST • 2 hrs (3 attempts every 33 hours)

Try again

Correct

Submitted Jul 14, 11:59 PM IST

That's correct! You can implement Gaussian blurring, box blurring and median filter blurring using GitHub Copilot.

2. What is the primary difference between Gaussian blurring and Box blurring in image processing?

5 / 5 points

100%

Gaussian blurring gives more weight to central pixels, whereas Box blurring treats all pixels equally.

☒

Gaussian blurring uses the median value, whereas Box blurring uses the average value of the pixels.

☐

Gaussian blurring gives equal weight to all pixels, whereas Box blurring gives more weight to central pixels.

☐

Gaussian blurring is faster than Box blurring.

☐

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Correct

Submitted Jul 14, 11:59 PM IST

Gaussian blurring calculates the average color value of each pixel and its neighborhood, giving more weight to the central pixels. In contrast, Box blurring treats all pixels equally, resulting in a uniform blur effect.

3. What is the key property of a binary search tree (BST)?

5 / 5 points

All elements in the right subtree are less than or equal to the node's value.

☐

Each node has at most three children.

☐

All elements in the left subtree are greater than the node's value.

☐

All elements in the left subtree are less than or equal to the node's value, and all elements in the right subtree are greater than the node's value.

☒

Correct

Submitted Jul 14, 11:59 PM IST

The key property of a binary search tree is that for each node, all elements in its left subtree are less than or equal to the node's value, and all elements in its right subtree are greater than the node's value. This property allows for efficient search, insertion, and deletion operations.

4. Which of the following describes the importance of maintaining a balanced structure in a binary search tree?

5 / 5 points

It makes the tree look aesthetically pleasing

☐

It ensures that the maximum number of levels in the tree is minimized

☒

It guarantees that each node will have exactly two children

☐

It allows the tree to store any type of data

☐

Correct

Submitted Jul 14, 11:59 PM IST

Correct! A balanced binary search tree ensures that the maximum number of levels (the height) is minimized. This results in more efficient search, insert, and delete operations.

5. What is the primary purpose of GitHub Copilot in problem-solving?

5 / 5 points

To provide suggestions and streamline the coding experience.

☒

To generate random solutions without considering the problem statement.

☐

To replace the need for human creativity and understanding.

☐

To introduce biases into code suggestions.

☐

Correct

Submitted Jul 14, 11:59 PM IST

GitHub Copilot's primary purpose in problem-solving is to provide suggestions and streamline the coding experience by offering helpful code snippets based on the context of the problem, ultimately assisting developers in writing code more efficiently.

6. Which of the following approaches does NOT correctly solve the two sum problem?

5 / 5 points

Using a tree

☒

Using sorting and binary search

☐

Brute force approach

☐

Using hash maps

☐

Correct

Submitted Jul 14, 11:59 PM IST

Using tree is not a correct way of solving the Two Sum problem. Please refer to the video titled "The Problem : Two Sum Problem in Data Structures and Algorithms" in week 3.

https://www.coursera.org/learn/programming-with-generative-ai/assignment/submission/Qhxy/graded-quiz-solving-miscellaneous-problems/view-feedback

1/1